An Unconditional Proof of the Unboundedness of OEIS A182134 via Maynard's Prime-Cluster Theorem

Expository Mathematical Note

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Abstract

Let p_n denote the n-th prime and let $a(n)$ be the number of primes in the open interval $(p_n, p_n^{1+1/n})$, as in OEIS A182134. We give a short unconditional proof that $\limsup_{n\to\infty} a(n) = \infty$. More precisely, for every fixed $m \geq 1$, the interval contains at least m primes for infinitely many n. The deduction combines Maynard's theorem on bounded clusters of arbitrarily many primes with the prime number theorem. We also record an immediate generalization to any interval $(p_n, p_n + F(n))$ with $F(n) \to \infty$. OEIS already identifies the unboundedness assertion as a consequence of Maier's theorem; the argument here is an alternative proof using a previously published result, not a new prime-gap theorem.

Keywords: prime clusters; prime gaps; unboundedness; A182134; prime number theorem.
OEIS entry: A182134.

1 Definition and statement

Write p_n for the n-th prime and let

$$a(n) = \#\{q \text{ prime} : p_n < q < p_n^{1+1/n}\}. \tag{1}$$

This is the sequence OEIS A182134 [1]. The inequalities are strict, as in its definition.

Theorem 1. *For every integer $m \geq 1$, there exist infinitely many integers n such that $a(n) \geq m$. In particular,*

$$\limsup_{n\to\infty} a(n) = \infty. \tag{2}$$

Proof. Fix $m \geq 1$. Maynard's bounded-prime-cluster theorem [2, Theorem 1.1] gives a constant $H_m < \infty$ such that

$$p_{n+m} - p_n \leq H_m \tag{3}$$

for infinitely many n.

We claim that the right endpoint in (1) moves a distance tending to infinity from p_n. Put

$$L_n = p_n^{1+1/n} - p_n = p_n \left(\exp\left(\frac{\log p_n}{n} \right) - 1 \right).$$

The prime number theorem gives $p_n \sim n \log n$ and thus $\log p_n \sim \log n$. In particular $\log p_n / n \to 0$, so $e^t - 1 \sim t$ as $t \to 0$ yields

$$L_n \sim \frac{p_n \log p_n}{n} \sim (\log n)^2 \longrightarrow \infty. \tag{4}$$

Hence $L_n > H_m$ for all sufficiently large n. Since (3) holds for infinitely many n, it also holds for infinitely many such sufficiently large n. For any of them,

$$p_n < p_{n+1} < \cdots < p_{n+m} \leq p_n + H_m < p_n + L_n = p_n^{1+1/n}.$$

Therefore the m distinct primes $p_{n+1}, \ldots, p_{n+m}$ all lie in the defining interval for $a(n)$, and $a(n) \geq m$. As m was arbitrary, (2) follows. □

2 Generalization to expanding prime-anchored intervals

Corollary 1. *Let $F : \mathbb{N} \to (0, \infty)$ satisfy $F(n) \to \infty$. Define*

$$b_F(n) = \#\{q \text{ prime} : p_n < q < p_n + F(n)\}.$$

Then for every $m \geq 1$ there are infinitely many n with $b_F(n) \geq m$, and therefore

$$\limsup_{n\to\infty} b_F(n) = \infty.$$

Proof. For fixed m, take H_m from (3). Since $F(n) \to \infty$, eventually $F(n) > H_m$. On each sufficiently large n satisfying (3), the primes $p_{n+1}, \ldots, p_{n+m}$ lie in $(p_n, p_n + F(n))$. □

Remark (Scope and priority). The result is unconditional: it uses neither the Riemann hypothesis nor any other unproved conjecture. The OEIS entry records the unboundedness conclusion as already established by an application of Maier's theorem (annotation dated 9 April 2015) [1]. The present proof is another elementary deduction from Maynard's *Small gaps between primes* [2]. No originality or priority is claimed for the unboundedness theorem. The separate OEIS comments about the possible rarity of small values of $a(n)$ are not addressed by this note.

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References

[1] T. Ordowski et al., *OEIS A182134: Number of primes p such that* $\text{prime}(n) < p < \text{prime}(n)^{1+1/n}$, The On-Line Encyclopedia of Integer Sequences. https://oeis.org/A182134 (accessed 25 October 2026).

[2] J. Maynard, *Small gaps between primes*, Annals of Mathematics **181** (2015), no. 1, 383–413. https://doi.org/10.4007/annals.2015.181.1.7.